

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 15

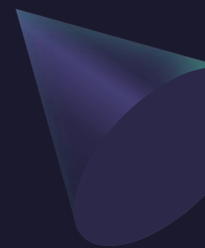


Any Reminders? Any Questions?

- Quiz on Friday on triple integrals in cartesian, cylindrical, and spherical coordinates.
- I will be away Wednesday-Thursday at a conference so no in-person class on 3/12 – watch videos instead!



ALVARO: Start the recording!



“Calculus 3”



Multi-Variable Calculus

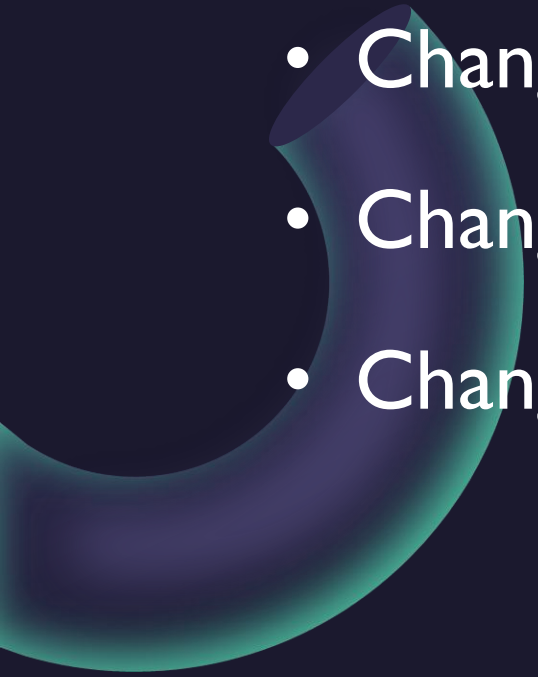
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Changes of Variables in Integrals





Today – Changes of Variables in Double and Triple Integrals

- 
- Changes of Variables in Single Integrals
 - Changes of Variables in Double Integrals
 - Changes of Variables in Triple Integrals

Changes of Variables in Single Integrals (aka u-Substitution)

$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$

$$\int_a^b f'(g(x)) g'(x) dx = \int_{a'}^{b'} f'(u) du = f(u) + C_1$$

$u = g(x)$
 $du = g'(x) dx$

//
 $= f(g(x)) + C$

no!

$$\int_a^b g'(x) dx = g(x) + C_1$$

Example: Evaluate the (single) integral

$$\int_0^2 x e^{x^2} dx$$

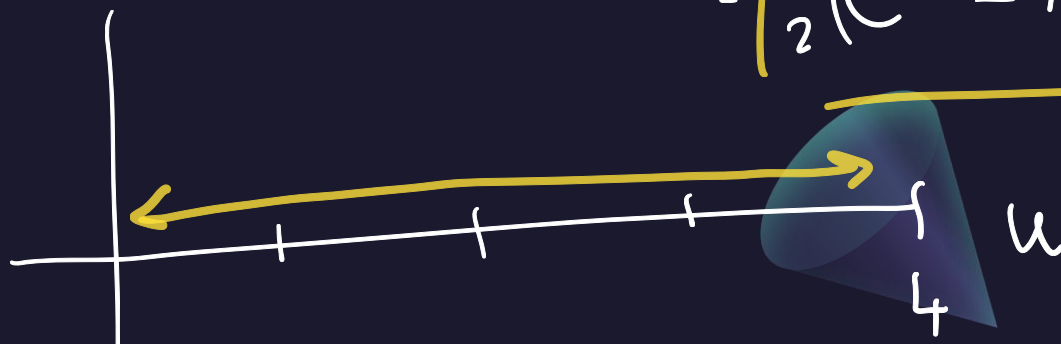
$$\int_0^2 x e^{x^2} dx =$$

$$= \int_0^4 e^u \frac{1}{2} du = \frac{1}{2} \int_0^4 e^u du = \frac{1}{2} e^u \Big|_0^4$$

$$= \frac{1}{2} (e^4 - 1)$$

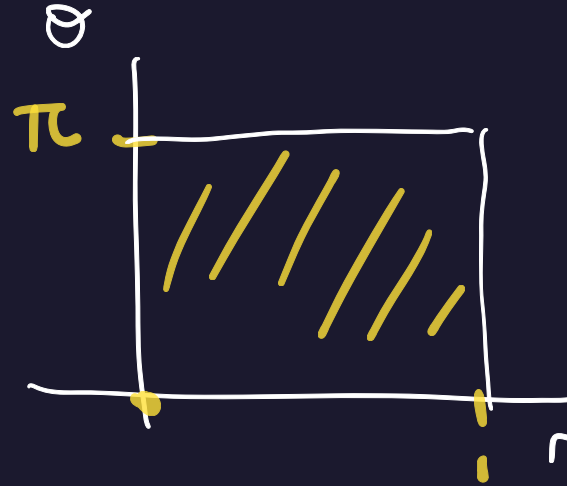
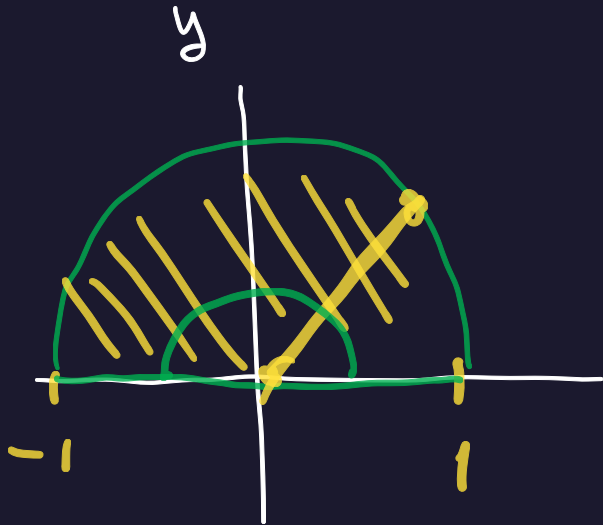
u
sub

$$\begin{cases} u = x^2 \\ du = 2x dx \\ x=0 \rightarrow u=0 \\ x=2 \rightarrow u=4 \end{cases} \quad x dx = \frac{1}{2} du$$



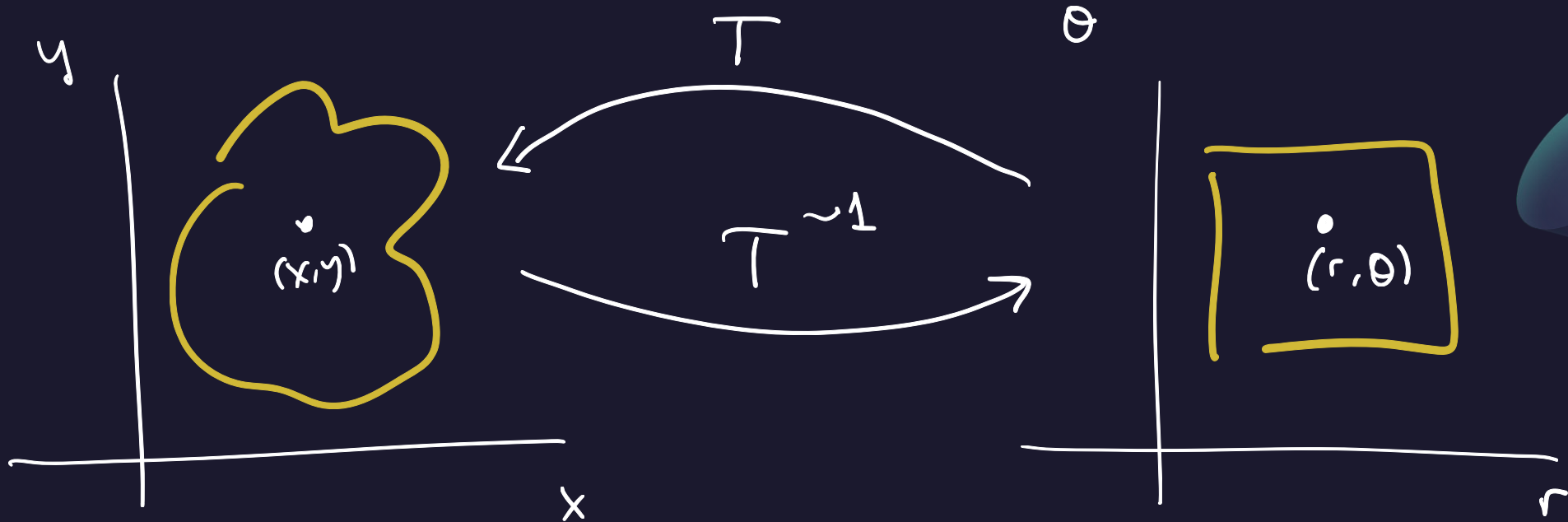
Changes of Variables in Double Integrals

Example: Evaluate the (double) integral $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} \sqrt{x^2 + y^2} \, dy \, dx$



$$I = \int_0^{\pi} \int_0^1 r \cdot r \, dr \, d\theta = \int_0^{\pi} \int_0^1 r^2 \, dr \, d\theta$$

Changes of Variables in Double Integrals (in Polar)



$$x = r \cos \theta$$
$$y = r \sin \theta$$

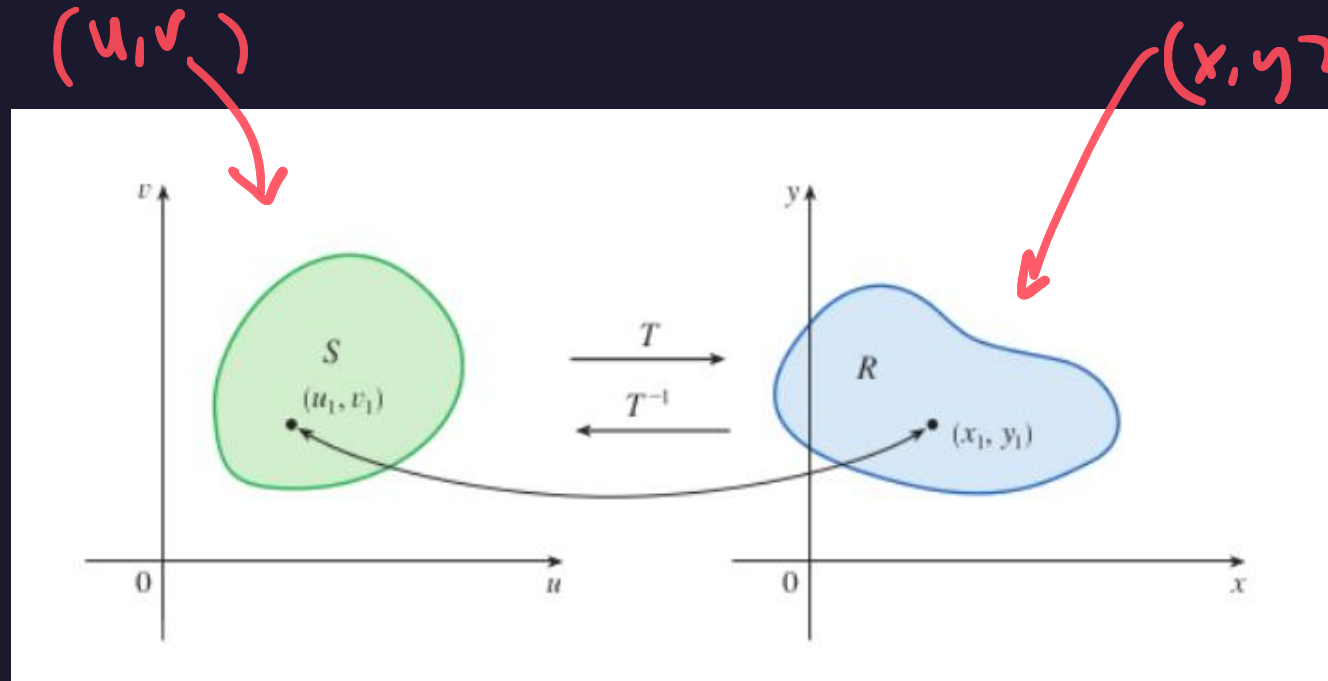
$$T(r, \theta) = (r \cos \theta, r \sin \theta)$$

$$T^{-1}(x, y) = (\sqrt{x^2 + y^2}, \tan^{-1} \frac{y}{x})$$

Changes of Variables in Double Integrals (in General)

$$T(u, v) = (f(u, v), g(u, v))$$

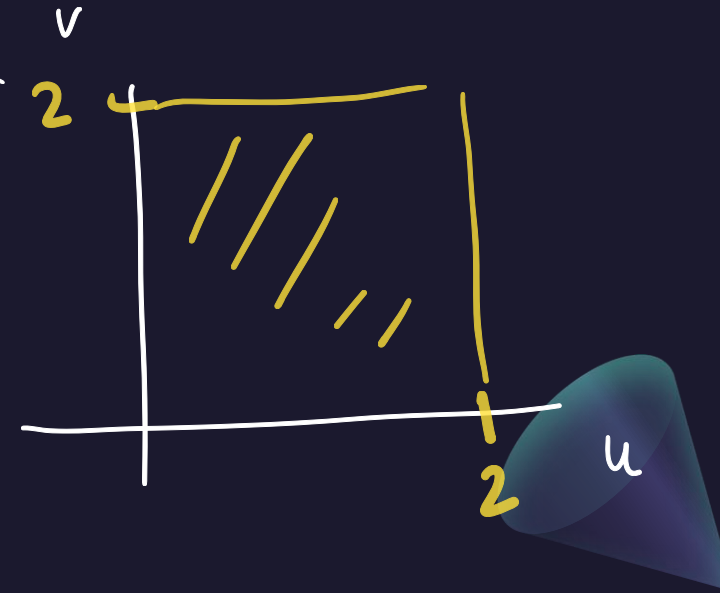
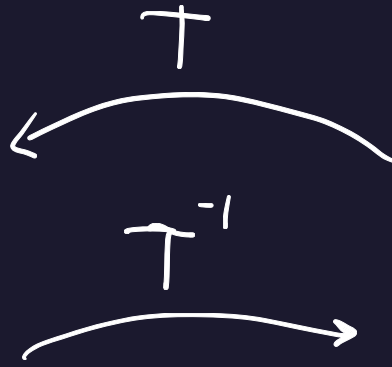
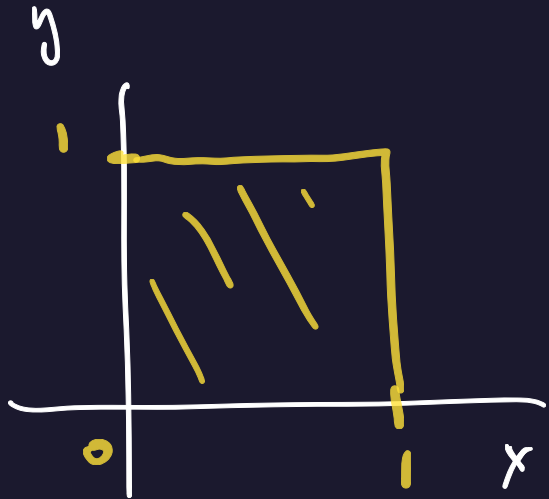
$$T^{-1}(x, y) = (h(x, y), k(x, y))$$



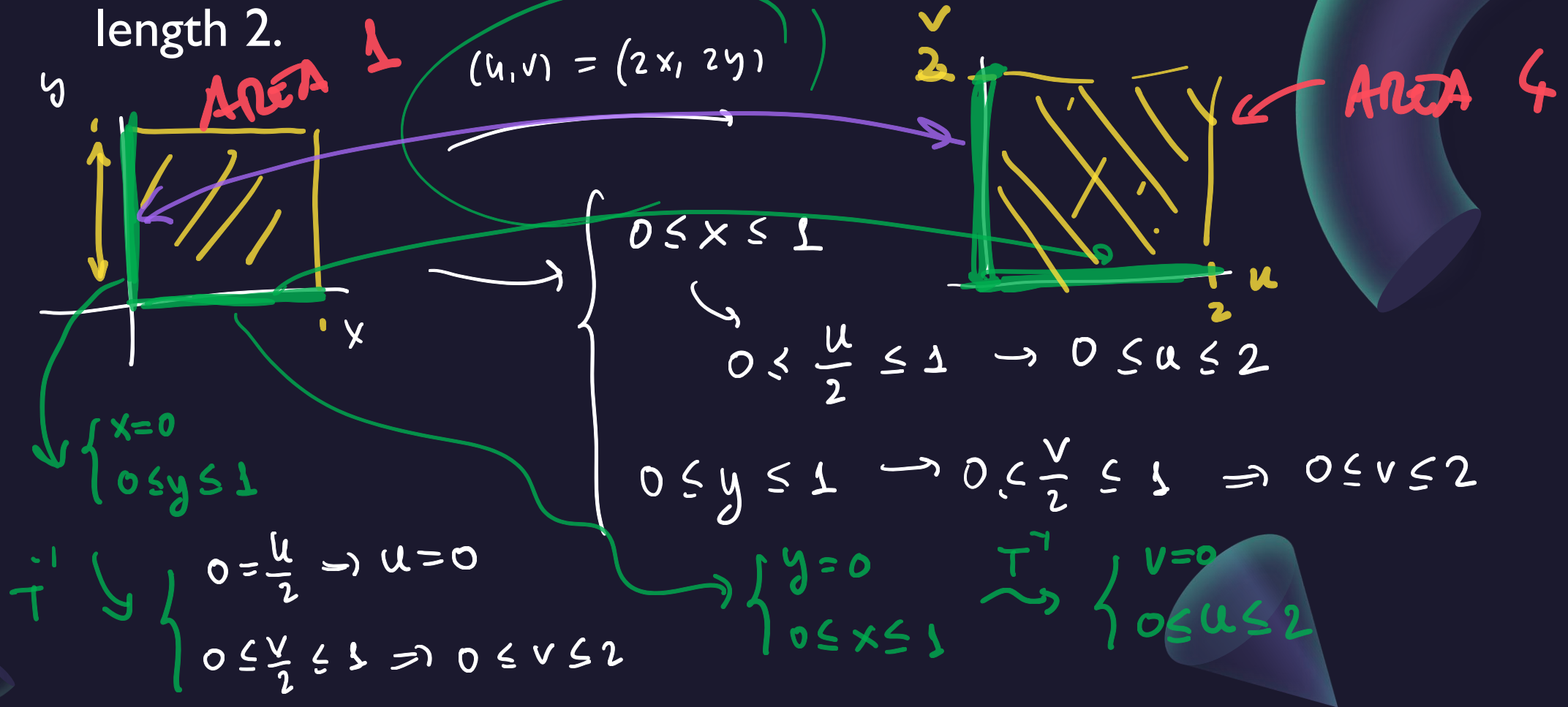
Example: Show that the change of variables $x = \frac{u}{2}, y = \frac{v}{2} \rightsquigarrow \begin{cases} u = 2x \\ v = 2y \end{cases}$ sends the unit square (in x, y coordinates, in the first quadrant) to a square of side length 2 (in u, v).

$$T(u, v) = \left(\frac{u}{2}, \frac{v}{2} \right)$$

$$T^{-1}(x, y) = (2x, 2y)$$



Example: Show that the change of variables $x = \frac{u}{2}, y = \frac{v}{2}$ sends the unit square (in the first quadrant) to a square of side length 2.



Example: Evaluate the (double) integrals

$$\underbrace{\int_0^1 \int_0^1 e^{2x+2y} dy dx}_{\substack{\text{1} \\ \boxed{1/4}}} \quad \text{and} \quad \underbrace{\int_0^2 \int_0^2 e^{u+v} dv du}_{\substack{\text{2} \\ \boxed{1/2}}} \quad \left| \begin{aligned} &= \left(\int_0^2 e^u du \right) \left(\int_0^2 e^v dv \right) \\ &= \boxed{(e^2 - 1)^2} \end{aligned} \right.$$

$\begin{cases} u = 2x \\ v = 2y \end{cases}$

$$\int_0^1 \int_0^1 e^{2x+2y} dy dx = \int_0^1 \int_0^1 e^{2x} \cdot e^{2y} dy dx$$

$$= \left(\int_0^1 e^{2x} dx \right) \cdot \left(\int_0^1 e^{2y} dy \right)$$

$$= \boxed{\frac{1}{4} (e^2 - 1)^2}$$

~~OFF~~

~~BY~~

~~1/4!~~

$$\begin{aligned} &\int_0^1 e^{2x} dx \\ &= \left. \frac{e^{2x}}{2} \right|_0^1 = \frac{e^2 - 1}{2} \end{aligned}$$

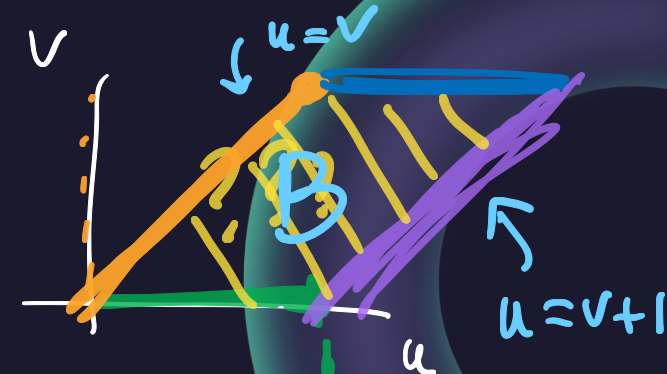
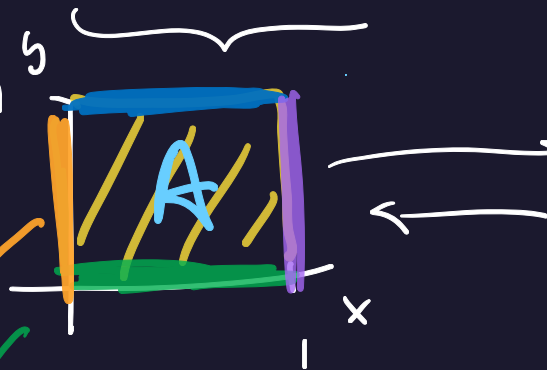
Example: Use the change of variables $x + y = u, y = v$ instead???

$$\int_0^1 \int_0^1 e^{2x+2y} dy dx$$

$$\begin{cases} x+y=u \\ y=v \end{cases} \longleftrightarrow \begin{cases} x=u-v \\ y=v \end{cases}$$

$$T(u,v) = (u-v, v)$$

$$T^{-1}(x,y) = (x+y, y)$$



(1) $\begin{cases} y=0 \\ 0 \leq x \leq 1 \end{cases}$

$\begin{cases} v=0 \\ u=x+y=x \\ 0 \leq u \leq 1 \end{cases}$

(2) $\begin{cases} x=0 \\ 0 \leq y \leq 1 \end{cases}$

$\begin{cases} u=x+y=y=v \\ 0 \leq v \leq 1 \end{cases}$

$$B = \begin{cases} v \leq u \leq v+1 \\ 0 \leq v \leq 1 \end{cases}$$

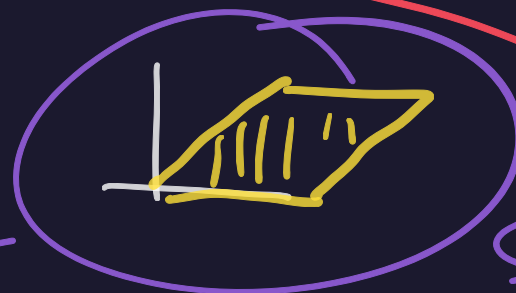
Example: Evaluate the (double) integrals

$$\int_0^1 \int_0^1 e^{2x+2y} dy dx \quad \text{and} \quad \int_0^1 \int_v^{v+1} e^{2u} du dv = ??$$

$$\frac{1}{4}(e^2-1)^2$$

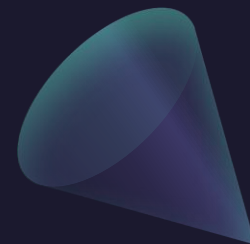


$$\begin{cases} u = x+y \\ v = y \end{cases}$$



OK!

SAME
AREA



$$\int_0^1 \int_v^{v+1} e^{2u} du dv = \int_0^1 \left. \frac{e^{2u}}{2} \right|_v^{v+1} dv$$

$$= \int_0^1 \frac{e^{2(v+1)}}{2} - \frac{e^{2v}}{2} dv = \left. \frac{e^{2(v+1)}}{4} - \frac{e^{2v}}{4} \right|_0^1$$

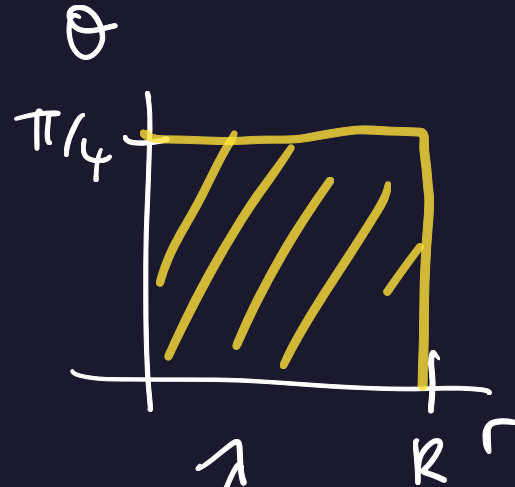
$$\frac{1}{4}(e^2-1)^2$$

$$= \frac{1}{4}(e^4 - e^2 - (e^2 - 1)) = \frac{1}{4}(e^4 - 2e^2 + 1)$$

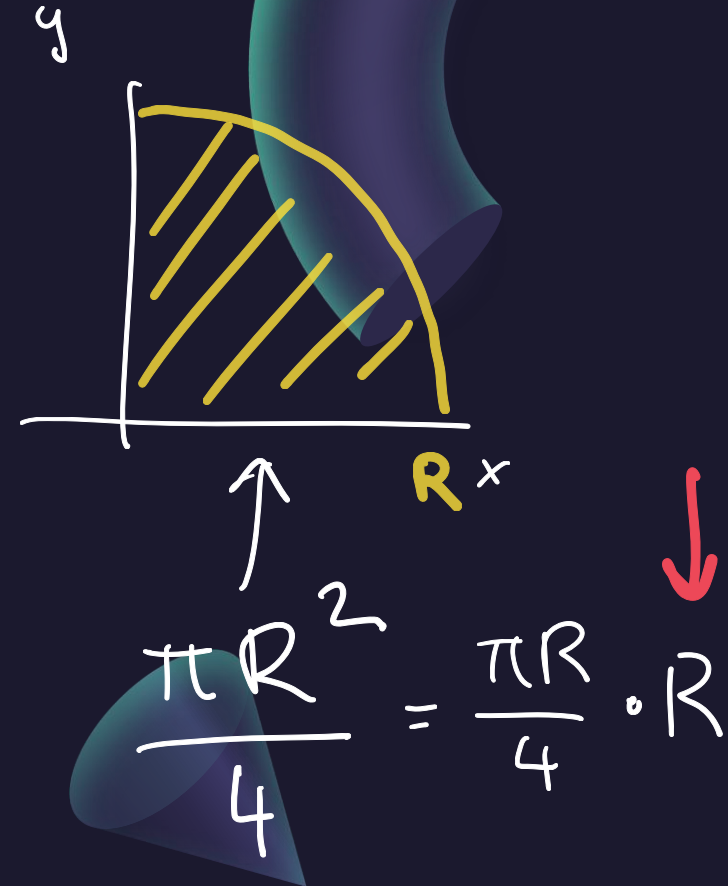


Example: Show that the change of variables $x = R \cos(\theta)$, $y = R \sin(\theta)$ sends the square of side lengths R and $\pi/4$ to a quarter circle of radius R . (Then, compute the areas of both regions.)

$$\begin{cases} x = R \cos \theta \\ y = R \sin \theta \end{cases}$$

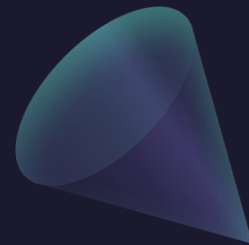


$$\text{Area } \frac{\pi}{4} \cdot R = \frac{\pi R}{4}$$

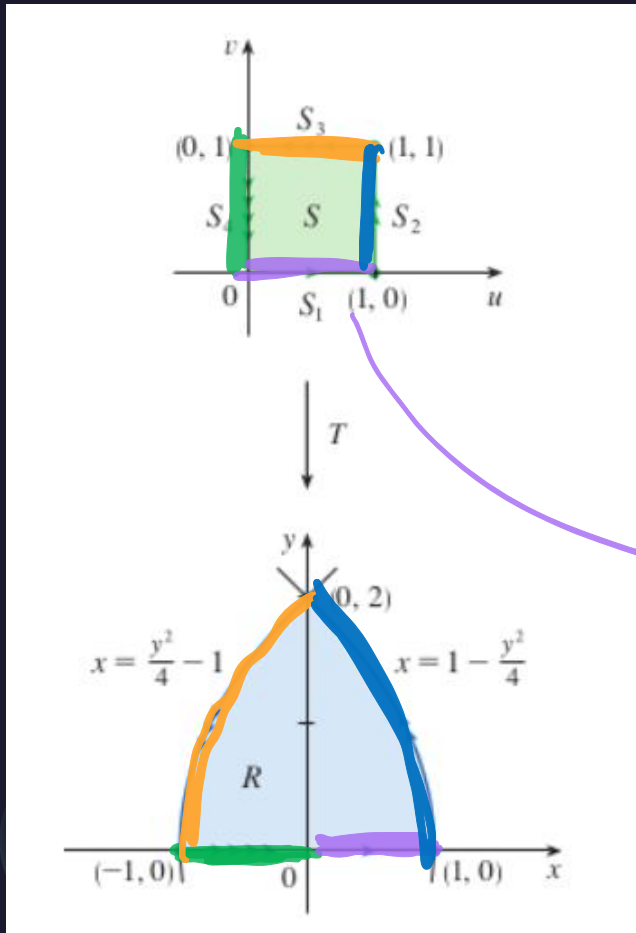


$$\frac{\pi R^2}{4} = \frac{\pi R}{4} \cdot R$$

Example: Show that the change of variables $x = R \cos(\theta)$, $y = R \sin(\theta)$ sends the square of side lengths R and $\pi/4$ to a quarter circle of radius R . (Then, compute the areas of both regions.)



Example: Show that the change of variables $x = u^2 - v^2, y = 2uv$ sends the region S to the region R , and vice-versa.



$$(1) \begin{cases} u=0 \\ 0 \leq v \leq 1 \end{cases}$$

$$\begin{cases} x = -v^2 \\ y = 0 \end{cases}$$

$$-1 \leq x \leq 0$$

$$(2) \begin{cases} 0 \leq u \leq 1 \\ v=1 \end{cases}$$

$$\begin{cases} x = u^2 - 1 \\ y = 2u \end{cases}$$

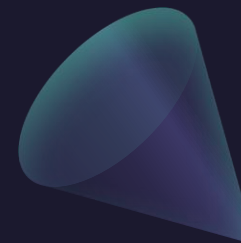
$$x = \frac{y^2}{4} - 1$$

$$u = y/2$$

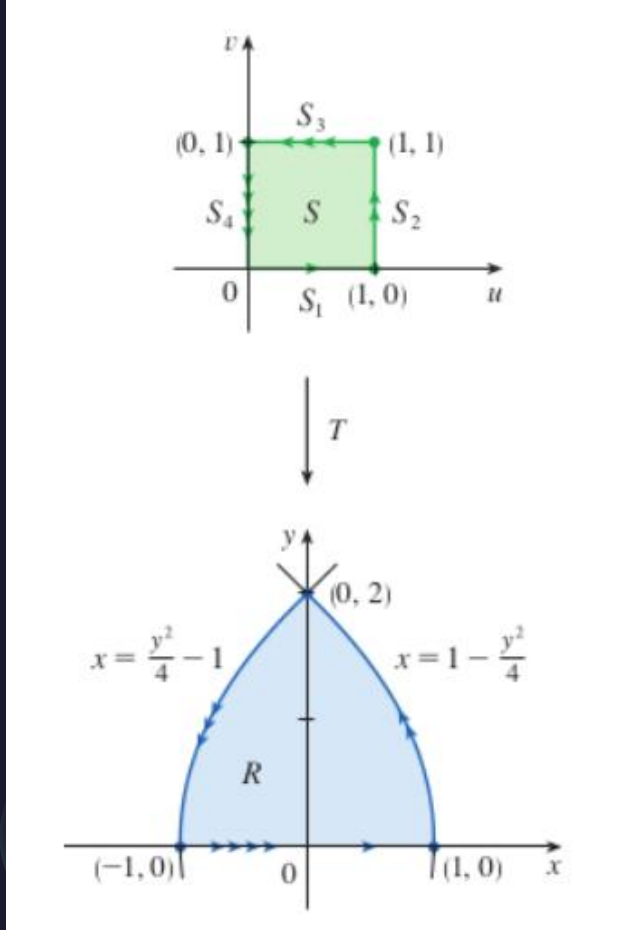
$$0 \leq y \leq 2$$

$$(3) \begin{cases} 0 \leq u \leq 1 \\ v=0 \end{cases}$$

$$\begin{cases} x = u^2 \\ y = 0 \end{cases}$$



Example: Show that the change of variables $x = u^2 - v^2, y = 2uv$ sends the region S to the region R , and vice-versa.

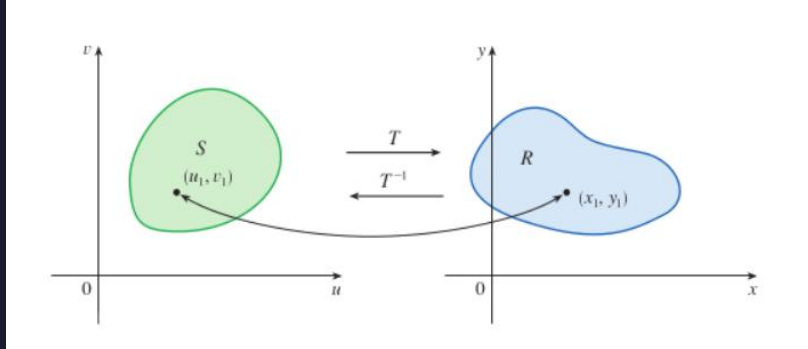


$$\int_0^1 \int_0^1 ? \, du \, dv$$

$$\int_0^2 \int_{\frac{y^2}{4} - 1}^{1 - \frac{y^2}{4}} dx \, dy$$



Changes of Variables in Double Integrals (in General)



CORRECTION
FACTOR

Suppose that T is a C^1 transformation whose Jacobian is nonzero and that T maps a region S in the uv -plane onto a region R in the xy -plane. Suppose that f is continuous on R and that R and S are type I or type II plane regions. Suppose also that T is one-to-one, except perhaps on the boundary of S . Then

$$\iint_R f(x, y) \, dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du \, dv$$

Changes of Variables in Double Integral: the Jacobian

$$\iint_R f(x, y) \, dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du \, dv$$

The **Jacobian** of the transformation T given by $x = g(u, v)$ and $y = h(u, v)$ is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}$$



The Jacobian is named after the German mathematician Carl Gustav Jacob Jacobi (1804–1851). Although the French mathematician Cauchy first used these special determinants involving partial derivatives, Jacobi developed them into a method for evaluating multiple integrals.

Example: Find the jacobian of the change of variables $x(R, \theta)$
 $x = R \cos(\theta), y = R \sin(\theta)$ $y(R, \theta)$

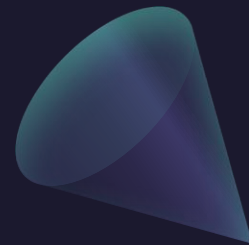
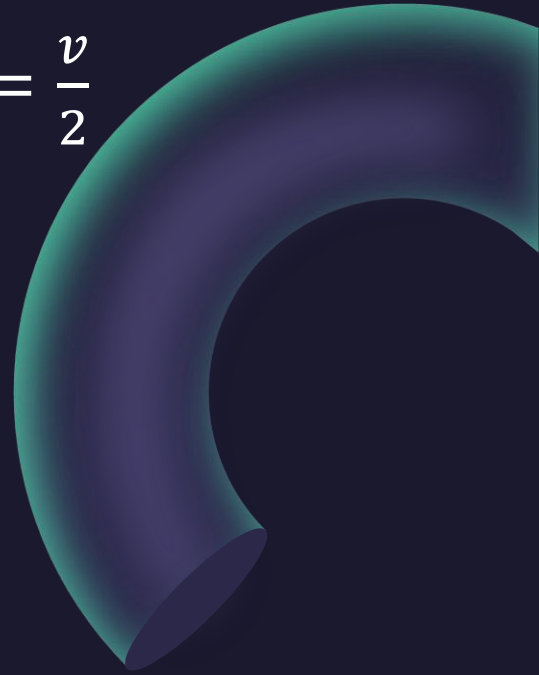
$$\text{Jacobian} = \begin{vmatrix} \frac{\partial x}{\partial R} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial R} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -R \sin \theta \\ \sin \theta & R \cos \theta \end{vmatrix}$$

$$= \cos \theta \cdot R \cos \theta - (\sin \theta \cdot (-R \sin \theta))$$

$$= R \cos^2 \theta + R \sin^2 \theta = R(\sin^2 \theta + \cos^2 \theta) = \boxed{R}$$

Example: Find the jacobian of the change of variables $x = \frac{u}{2}, y = \frac{v}{2}$

$$J_{ac} = \begin{vmatrix} 1/2 & 0 \\ 0 & 1/2 \end{vmatrix} = \frac{1}{2} \cdot \frac{1}{2} - 0 \cdot 0 =$$
$$= \boxed{1/4}$$



Example: Find the jacobian of the change of variables

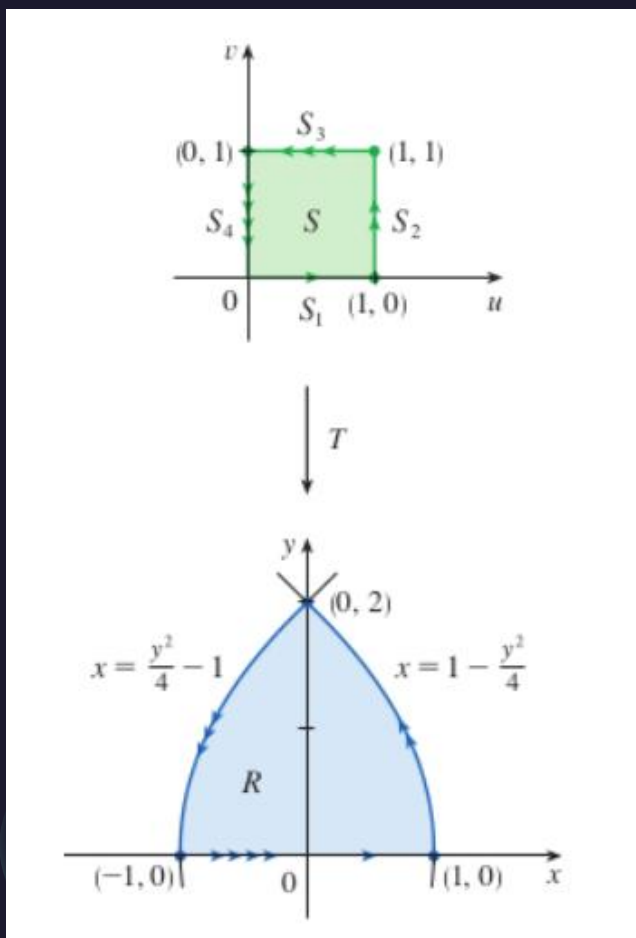
$$x + y = u, y = v$$

$$\begin{cases} x + y = u \\ y = v \end{cases} \quad \leftarrow \quad \begin{cases} x = u - y = u - v \\ y = v \end{cases} \quad \leftarrow \quad \begin{cases} x = u - v \\ y = v \end{cases}$$

$$J_{ac} = \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} = 1 \cdot 1 - (-1) \cdot 0 = \boxed{1}$$

Example: Find the jacobian of the change of variables

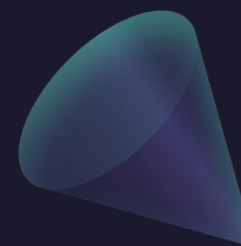
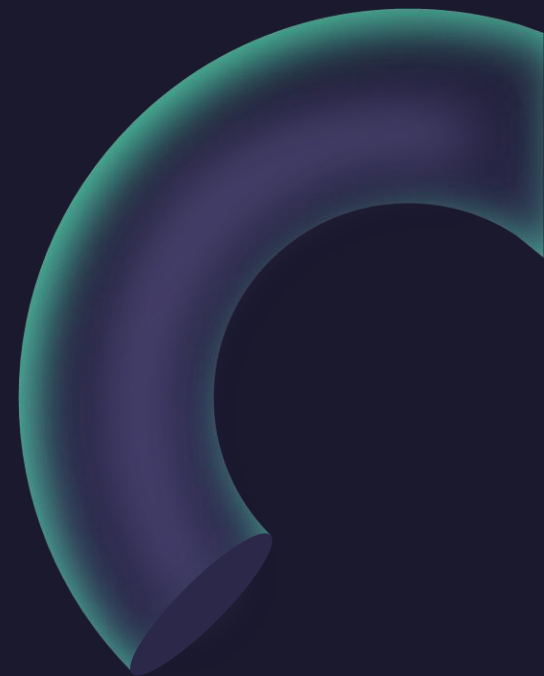
$$x = u^2 - v^2, y = 2uv$$



$$J_{ac} = \begin{vmatrix} 2u & -2v \\ 2v & 2u \end{vmatrix}$$

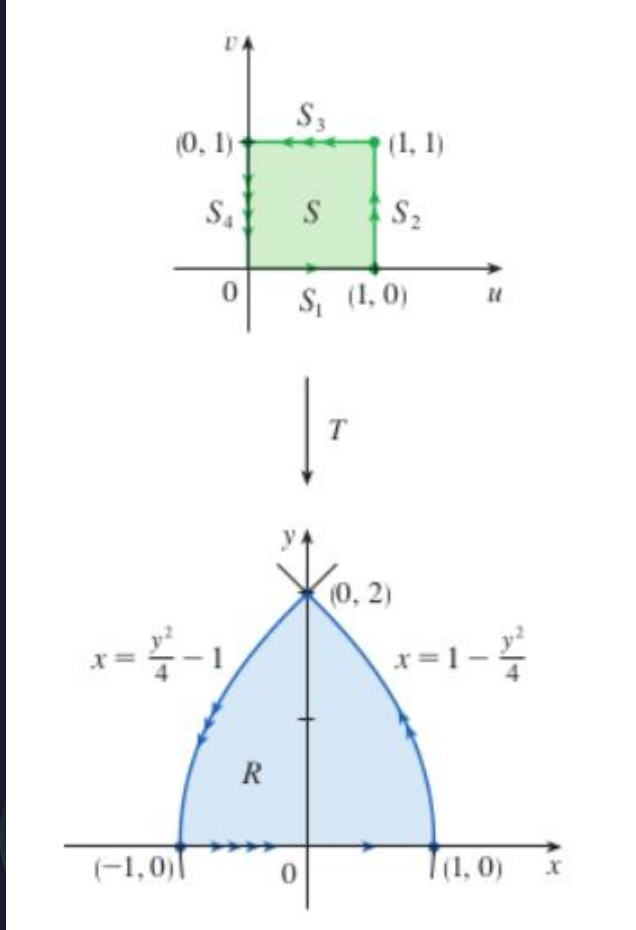
$$= 4u^2 + 4v^2$$

$$= 4(u^2 + v^2)$$



Example: Use the change of variables $x = u^2 - v^2$, $y = 2uv$ to evaluate $\iint_R y \, dA$

where R is the region bounded by the parabolas $y^2 = 4 - 4x$ and $y^2 = 4 + 4x$ and $y \geq 0$.



$$\int_0^2 \int_{\frac{y^2}{4} - 1}^{1 - \frac{y^2}{4}} y \, dx \, dy$$

correc. factor = jacobian

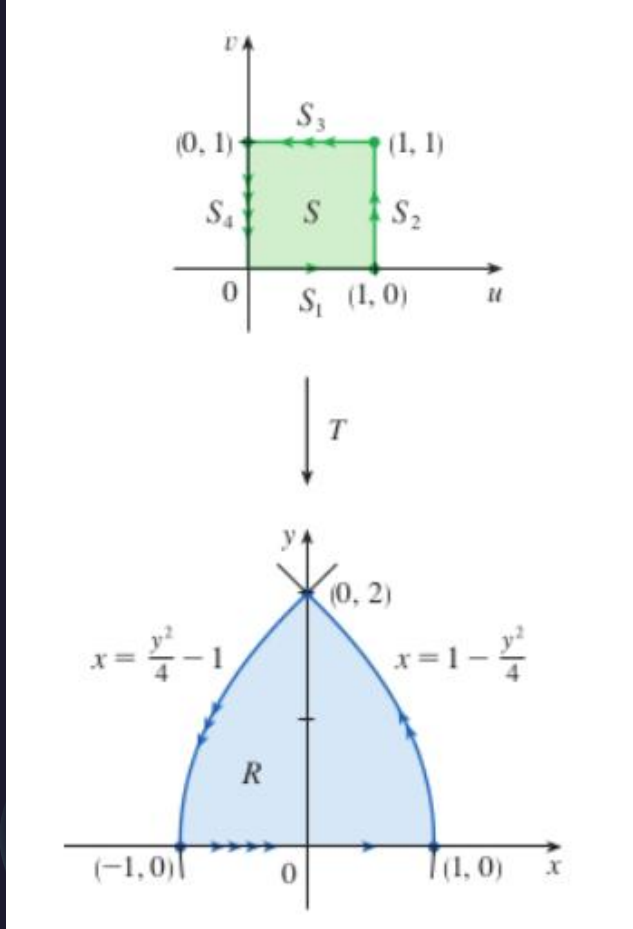
$$= \int_0^1 \int_0^1 2uv \cdot 4(u^2 + v^2) \, du \, dv$$

$$= 2$$

Example: Use the change of variables $x = u^2 - v^2, y = 2uv$ to evaluate

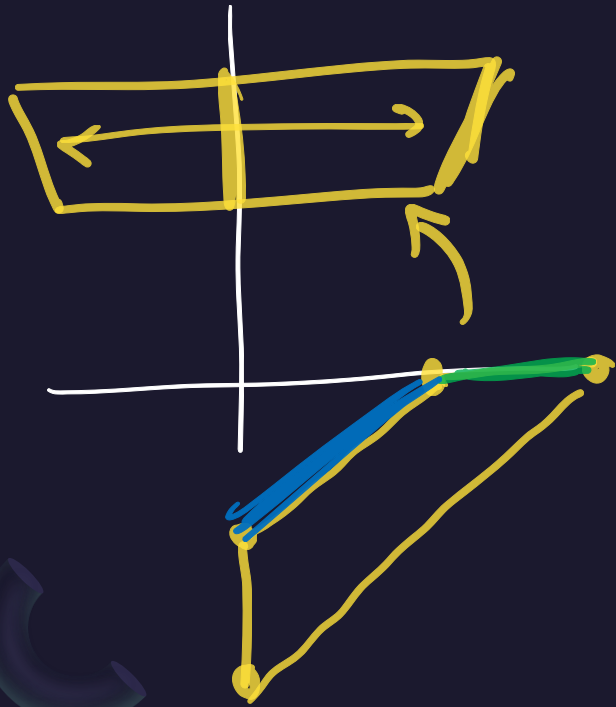
$$\iint_R y \, dA$$

where R is the region bounded by the parabolas $y^2 = 4 - 4x$ and $y^2 = 4 + 4x$ and $y \geq 0$.



Example: Evaluate the integral $\iint_R e^{\frac{x+y}{x-y}} dA$ $\xrightarrow{\quad} e^{u/v}$

where R is the trapezoidal region with vertices $(1,0)$, $(2,0)$, $(0,-2)$, $(0,-1)$.



$$\begin{cases} u = x+y \\ v = x-y \end{cases}$$

$$\longleftrightarrow \begin{cases} x = \frac{u+v}{2} \\ y = \frac{u-v}{2} \end{cases}$$

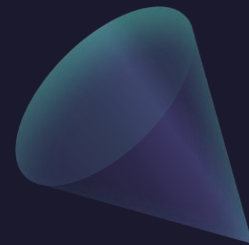
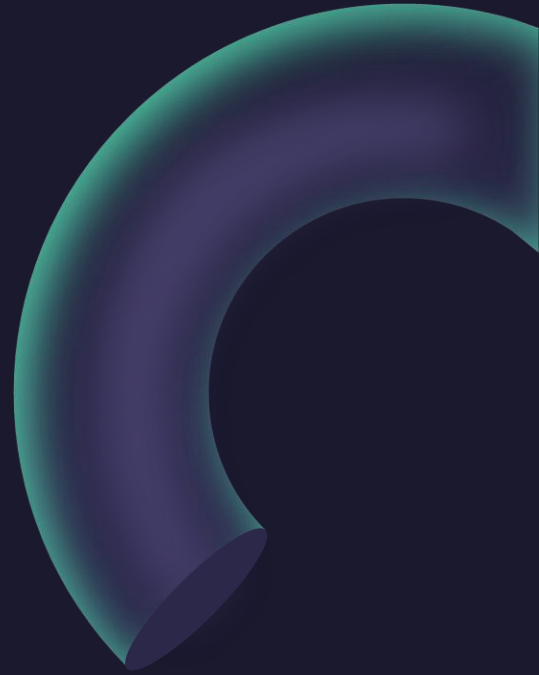
$$Jac = \begin{vmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{vmatrix}$$

$$\begin{aligned} y &= u - x \\ v &= x - (u - x) \\ v + u &= 2x \\ x &= \frac{u+v}{2} \end{aligned}$$

$$\begin{aligned} y &= u - x \\ &= u - \frac{u+v}{2} \\ &= \frac{u}{2} - \frac{v}{2} = \frac{u-v}{2} \end{aligned}$$

$$= \frac{1}{4} - \frac{1}{4} = \boxed{-\frac{1}{2}}$$

Example: Evaluate the integral $\iint_R e^{\frac{x+y}{x-y}} dA$
where R is the trapezoidal region with vertices
 $(1,0)$, $(2,0)$, $(0,-2)$, $(0,-1)$.



Example:

Use the change of variables $x = 2u, y = 3v$ to evaluate the integral

$$\iint_R x^2 dA,$$

where R is region bounded by ellipse $9x^2 + 4y^2 = 36$.

Example:

Use the change of variables $x = 2u, y = 3v$ to evaluate the integral

$$\iint_R x^2 dA,$$

where R is region bounded by ellipse $9x^2 + 4y^2 = 36$.

Example:

Use the change of variables to evaluate the integral

$$\iint_R \sin(4x^2 + y^2) dA,$$

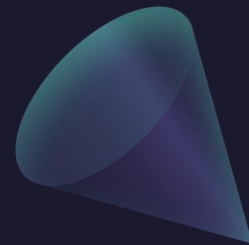
where R is the region in the first quadrant bounded by ellipse $4x^2 + y^2 = 1$.

Example:

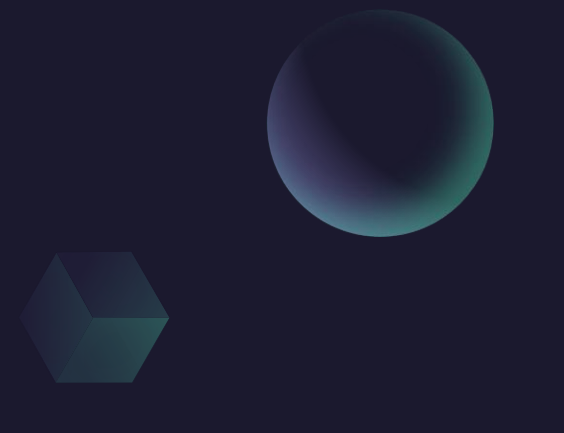
Use the change of variables to evaluate the integral

$$\iint_R \sin(4x^2 + y^2) dA,$$

where R is the region in the first quadrant bounded by ellipse $4x^2 + y^2 = 1$.



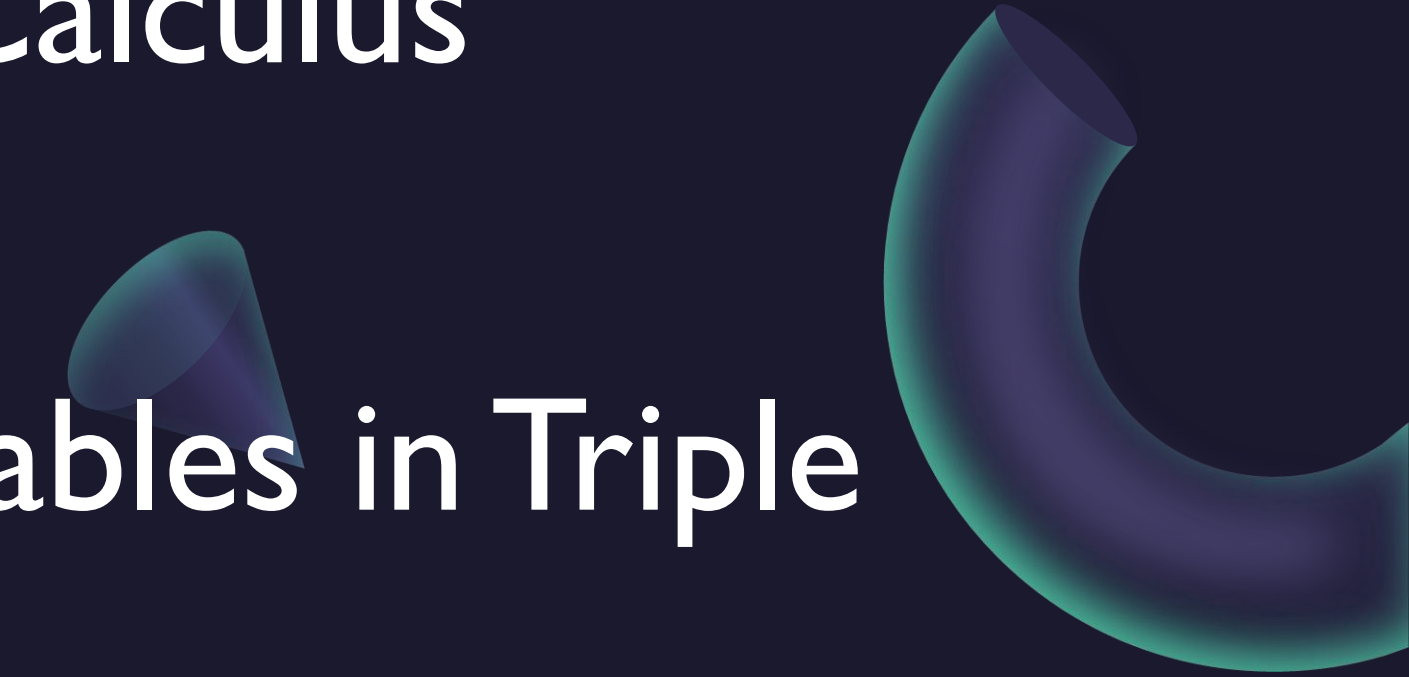
“Calculus 3”

A sphere, a cube, and a cone are positioned in the upper right area of the slide. The sphere is at the top right, the cube is below it and to the left, and the cone is further down and to the left.

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Changes of Variables in Triple Integrals

A cone and a torus are positioned in the lower right area of the slide. The cone is located above the word 'Integrals', and the torus is to the right of the word 'Triple'.

Changes of Variables in Triple Integrals (in General)

$$\iiint_R f(x, y, z) \, dV = \iiint_S f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du \, dv \, dw$$

$$x = g(u, v, w) \quad y = h(u, v, w) \quad z = k(u, v, w)$$

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

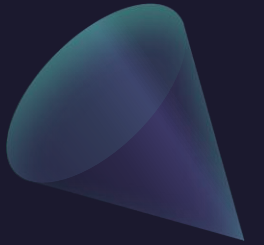
Example: Spherical Coordinates

$$x = \rho \sin \phi \cos \theta \quad y = \rho \sin \phi \sin \theta \quad z = \rho \cos \phi$$

We compute the Jacobian as follows:

$$\begin{aligned} \frac{\partial(x, y, z)}{\partial(\rho, \theta, \phi)} &= \begin{vmatrix} \sin \phi \cos \theta & -\rho \sin \phi \sin \theta & \rho \cos \phi \cos \theta \\ \sin \phi \sin \theta & \rho \sin \phi \cos \theta & \rho \cos \phi \sin \theta \\ \cos \phi & 0 & -\rho \sin \phi \end{vmatrix} \\ &= \cos \phi \begin{vmatrix} -\rho \sin \phi \sin \theta & \rho \cos \phi \cos \theta \\ \rho \sin \phi \cos \theta & \rho \cos \phi \sin \theta \end{vmatrix} - \rho \sin \phi \begin{vmatrix} \sin \phi \cos \theta & -\rho \sin \phi \sin \theta \\ \sin \phi \sin \theta & \rho \sin \phi \cos \theta \end{vmatrix} \\ &= \cos \phi (-\rho^2 \sin \phi \cos \phi \sin^2 \theta - \rho^2 \sin \phi \cos \phi \cos^2 \theta) \\ &\quad - \rho \sin \phi (\rho \sin^2 \phi \cos^2 \theta + \rho \sin^2 \phi \sin^2 \theta) \\ &= -\rho^2 \sin \phi \cos^2 \phi - \rho^2 \sin \phi \sin^2 \phi = -\rho^2 \sin \phi \end{aligned}$$

Questions?



Thank you

Until next time.

